BT102 Assignment solution2

Last Date :19-july 2021

Subscribe for more solutions at

https://www.youtube.com/channel/UCh-swQpEKauZh4oSbk1z3Yg/featured

Download all ZOO, BT, BIO, CS, BIF Files at Website

Zainab Arshad Official

Questions:

(A) Explain the bacterial growth curve and how can we estimate the microbial growth?

Growth Curve

Bacterial growth curve: A **curve** on a graph that shows the changes in size of a **bacterial** population over time in a **culture**. ... Samples are removed at intervals and the number of viable **bacteria** is counted. A logarithmic **growth curve** is plotted, which shows various phases.

Bacterial colonies progress through **four phases** of **growth**: the lag **phase**, the log **phase**, the stationary **phase**, and the death **phase**. The generation time, which varies among bacteria, is controlled by many environmental conditions and by the nature of the **bacterial** species.



Estimation of the microbial growth

The rate of exponential **growth** of a **bacterial** culture is expressed as generation time, also the doubling time of the **bacterial** population. Generation time (G) is defined as the time (t) per generation (n = number of generations). Hence, G=t/n is the **equation** from which **calculations** of generation time (below) derive

Q2. **What are the growth requirements of microbes, discuss briefly.**

Zainab Arsl

Growth Requirements

Physical growth requirements of microbes

1. Temperature
2. . pH 3.
3. Osmotic Pressure

Chemical:

1) Source of C 2. N, S, P, O,

2) Trace elements

3) Growth factors

Physical factors

1) Temperature

Every group has Minimum growth temp Optimum G. T and Maximum G. T. the Min and max temp is 300C apart

Psychrotrophs bacteria Grow best at 20 – 30 0C and also Can grow at refrigeration temp They Spoil foods

Hyperthermophiles bacteria Optimum temp is 80 0C they Archaea in nature and Live in hot springs

2) **pH (-log of H+)** Enzymes need certain pH for optimal function. Most grow at pH between 6.5 and 7.5 Effect of pH it Very few grow below pH 4 Acid is produced by bacteria during fermentation

Acidophiles : Molds and yeast grow At pH of 5 to 6 Alkalinity also inhibits microbial growth –

Zainab Arshad Official

Rarely used for preservation concentrations for growth . the Obligate halophiles 30 % salt • Extreme halophiles Require high salt concentrations for growth the Facultative halophiles • 2% Salt

3)Osmotic pressure:

Removal of water and addition of salt to meat creates a solute-rich environment where **osmotic pressure** draws water out of microorganisms, thereby retarding their **growth**.

Chemical factors:

1) Carbon, S, P requirements:

Besides water, C is the most important requirement. It is Structural backbone for living matter all organic compounds has it. Half the dry wt of microbes is C. Chemoheterotrophs Use C from energy sources Proteins, carbs, and lipids Chemoautotrophs and photoautotrophs Use CO2 N, S, and P Nitrogen and Sulfur needed for protein syntheis . Nitrogen and phosphorus

Zainab Arsha Official

needed for DNA and RNA synthesis and ATP

How bacteria get

2)**Oxygen requirement**

Needed for aerobic respiration

Use of O2 means more ATP

NADH and FADH2 during Kreb's cycle oxidized via ETC and generate 36 or 38 ATP molecules the Obligate Aerobes Need O2 for growth like Mycobacterium tuberculosis and Facultative Anaerobes for example E. coli

3)**trace elements** :

The **trace elements** usually act as cofactors for essential enzymatic reactions in the cell. One organism's **trace element** may be another's required **element** and vice-versa, but the usual cations that qualify as **trace elements** in **bacterial** nutrition are Mn, Co, Zn, Cu, and Mo.

4) Agar for solidification

Shad Official

The Derived from a sea weed . The Complex carb, not used by bacteria. it Liquefies at 100 0C and solidifies at 4o 0C – Petri plates – Slants

The following types Media is requires for bacterial growth

1)Blood Agar: it is a Nutrient agar + blood

2)MacConkey Agar

3) Nutrient agar + Bile salt + CV

4)Pure Culture

(B) A pastry chef accidentally inoculated a cream pie with six S. aureus cells. If S. aureus has a generation time of 60 minutes, how many cells would be in the cream pie after 7 hours?

Answer:

Generation time=60 minutes= 1 hour

This means in 1 hour 1 cell divides into 2.

Therefore,number of cells after 7 hours will be 2^7=128 cells.

Shahnab Arshad Official